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Build a desktop app with Qt and Rust

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Azzam crafts software for people to enjoy. Azzam is an OSS devotee, speaker, and teacher.

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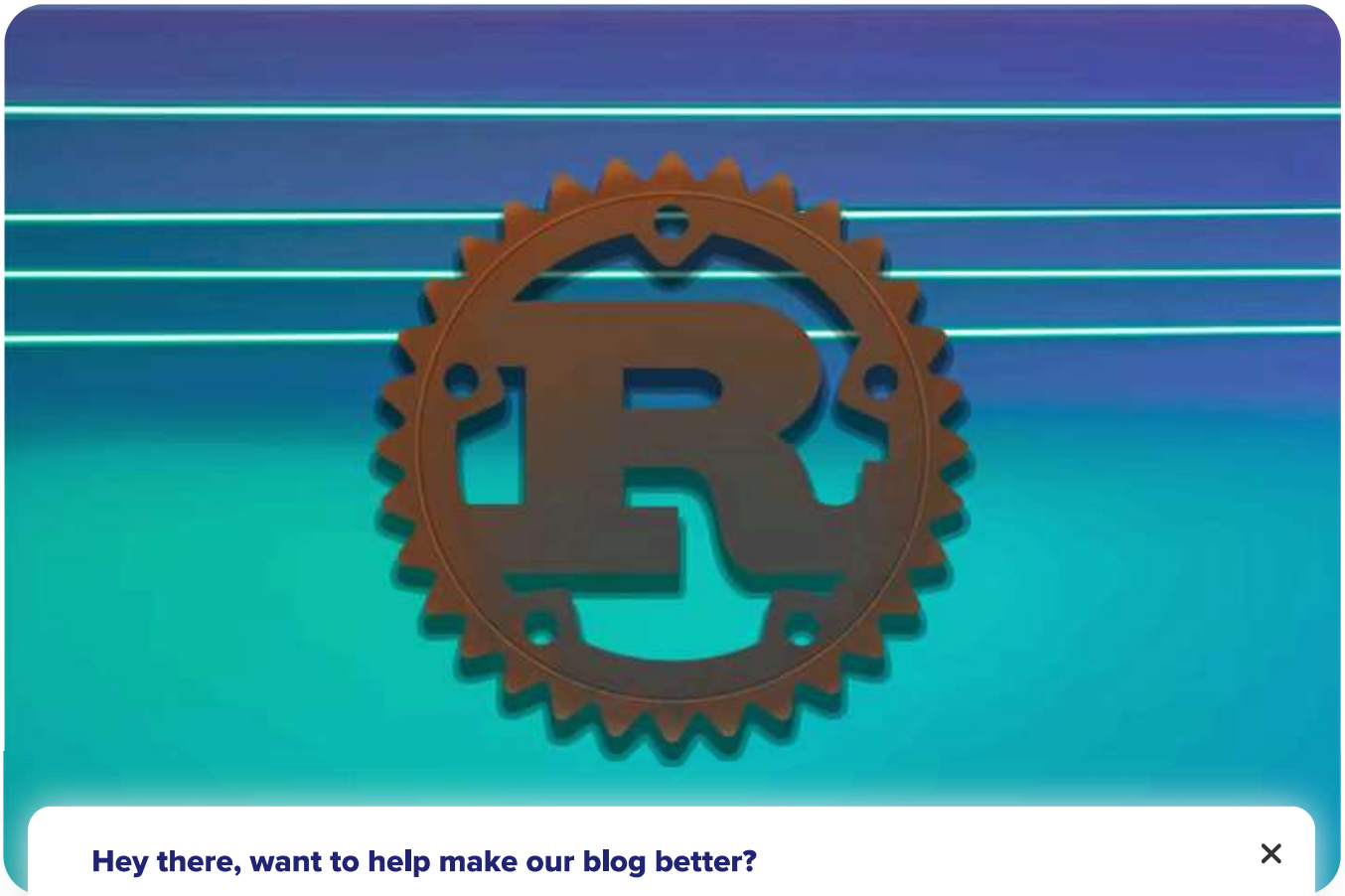
The rise of progressive web applications has resulted in a commensurate increase in the number of desktop apps that are released every day. To see evidence of this, just go to Flathub's new apps page or GitHub's trending page. For example, immediately after the release of the ChatGPT API, hundreds of desktop applications emerged. Desktop apps are native, fast, and secure and provide experiences that web applications can't match.



cmOf all the programming languages used in desktop app development, Rust is one of the more popular. Rust is widely regarded as being reliable, performant, productive, and versatile. In fact, many organizations are migrating their applications to Rust. The GNOME Linux development environment is an example. Personally, I especially love Rust's reliability principle: "If it compiles, it works."

In this article, we'll demonstrate how to build a desktop application using Qt (a mature, battle-tested framework for cross-platform development) and Rust.

Jump ahead:

Prerequisites

T
h

Choosing Rust Qt bindings

Rust has several Qt bindings. The most popular are Ritual, CXX-Qt, and `qmetaobject`. Ritual is not maintained anymore, and `qmetaobject` doesn't support `QWidgets`. So CXX-Qt is our best bet for now.

Because Rust is a relatively new language. So is the ecosystem. CXX-Qt is not as mature as PyQt. But it is on the way there. The current latest release already has a good and simple API.

Getting started with Rust and Qt

Install Rust using the following command:

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
```

If you're using Windows, go to https://rustup.rs/ to download the installer. To ensure everything is okay, run the following command in your terminal:

```
rustc --version
```

Next, install Qt:

```
# Ubuntu
sudo apt install qt6-base-dev qt6-declarative-dev

# Fedora
sudo dnf install qt6-qtbase-devel qt6-qtdeclarative-devel

# If you are unsure, just install all Qt dependencies
```

```
sudo apt install qt6*
sudo dnf install qt6*
```

To check that Qt is successfully installed, check that you're able to run the following:

```
qmake --version
```

Everything looks great! We are good to go now.

Application components

CXX-Qt is the Rust Qt binding. It provides a safe mechanism for bridging between Qt code and Rust. Unlike typical one-to-one bindings. CXX-Qt uses CXX to bridge Qt and Rust. This gives more powerful code, safe API, and safe multi-threading between both codes. Unlike the previous version. You don't need to touch any C++ code in the latest version.

QML is a programming language to develop the user interface. It is very readable because it offers JSON-like syntax. QML also has support for imperative JavaScript expressions and dynamic property bindings; this will be helpful for writing our Caesar Cipher application. If you need a refresher, see this QML intro.

Qt application demo

To demonstrate how to work with Qt and Rust, we'll build a simple "Hello World" application.

Creating the Rust project

F

```
> cargo new --bin demo
     Created binary (application) `demo` package
```

Next, open the `Cargo.toml` file and add the dependencies:

```
[dependencies]
cxx = "1.0.83"
cxx-qt = "0.5"
cxx-qt-lib = "0.5"

[build-dependencies]
cxx-qt-build = "0.5"
```

Now, let's create the entry point for our application. In the `src/main.rs` file we'll initialize the GUI application and the QML engine. Then we'll load the QML file and tell the application to start:

```
// src/main.rs

use cxx_qt_lib::{QGuiApplication, QQmlApplicationEngine, QUrl};

fn main() {
    // Create the application and engine
    let mut app = QGuiApplication::new();
    let mut engine = QQmlApplicationEngine::new();

    // Load the QML path into the engine
    if let Some(engine) = engine.as_mut() {
        engine.load(&QUrl::from("qrc:/main.qml"));
    }

    // Start the app
```

```
    }
}
```

To set up communication between Rust and Qt, we'll define the object in the `src/cxxqt_oject.rs` file:

```
// src/cxxqt_object.rs

#[cxx_qt::bridge]
mod my_object {

    #[cxx_qt::qobject(qml_uri = "demo", qml_version = "1.0")]
    #[derive(Default)]
    pub struct Hello {}

    impl qobject::Hello {
        #[qinvokable]
        pub fn say_hello(&self) {
            println!("Hello world!")
        }
    }
}
```

Attribute macro is used to enable CXX-Qt features.

- `#[cxx_qt::bridge]` : marks the Rust module to be able to interact with C++
- `#[cxx_qt::qobject]` : expose a Rust struct to Qt as a QObject subclass
- `#[qinvokable]` : expose a function on the QObject to QML and C++ as a Q_INVOKABLE.

Next, we'll create a `struct` , named `Hello` , derived from the `qobject` traits. Then we can implement our regular Rust function to print a greeting:

```
+ mod cxxqt_object;

use cxx_qt_lib::{QGuiApplication, QQmlApplicationEngine, QUrl};
```

N.B., *don't forget to tell Rust if you have a new file*

Designing the UI

We'll use QML to design the user interface. The UI file is located in the `qml/main.qml` file:

```
// qml/main.qml

import QtQuick.Controls 2.12
import QtQuick.Window 2.12

// This must match the qml_uri and qml_version
// specified with the #[cxx_qt::qobject] macro in Rust.
import demo 1.0

Window {
    title: qsTr("Hello App")
    visible: true
    height: 480
    width: 640
    color: "#e4af79"

    Hello {
        id: hello
    }

    Column {
```

```
        Button {
            text: "Say Hello!"
            onClicked: hello.sayHello()
        }
    }
}
```

If you look closely at `sayHello` function, you'll notice that CXX-Qt converts the Rust function's snake case to the camelCase C++ convention. Now our QML code doesn't look out of place!

Next, we have to tell Qt about the QML location using the Qt resource file. It should be located in the `qml/qml.qrc` file:

```
<!DOCTYPE RCC>
<RCC version="1.0">
    <qresource prefix="/">
        <file>main.qml</file>
    </qresource>
</RCC>
```

Building the application

The last step is to build the application. To teach Rust how to build the `cxxqt_object.rs` and QML files, we need to first define it in `build.rs` file:

```
// build.rs

use cxx_qt_build::CxxQtBuilder;
```

```
        // - Qt Gui is linked by enabling the qt_gui Cargo feature (d
        // - Qt Qml is linked by enabling the qt_qml Cargo feature (d
        // - Qt Qml requires linking Qt Network on macOS
        .qt_module("Network")
        // Generate C++ from the `#[cxx_qt::bridge]` module
        .file("src/cxxqt_object.rs")
        // Generate C++ code from the .qrc file with the rcc tool
        // https://doc.qt.io/qt-6/resources.html
        .qrc("qml/qml.qrc")
        .setup_linker()
        .build();
}
```

The final structure should look like this:

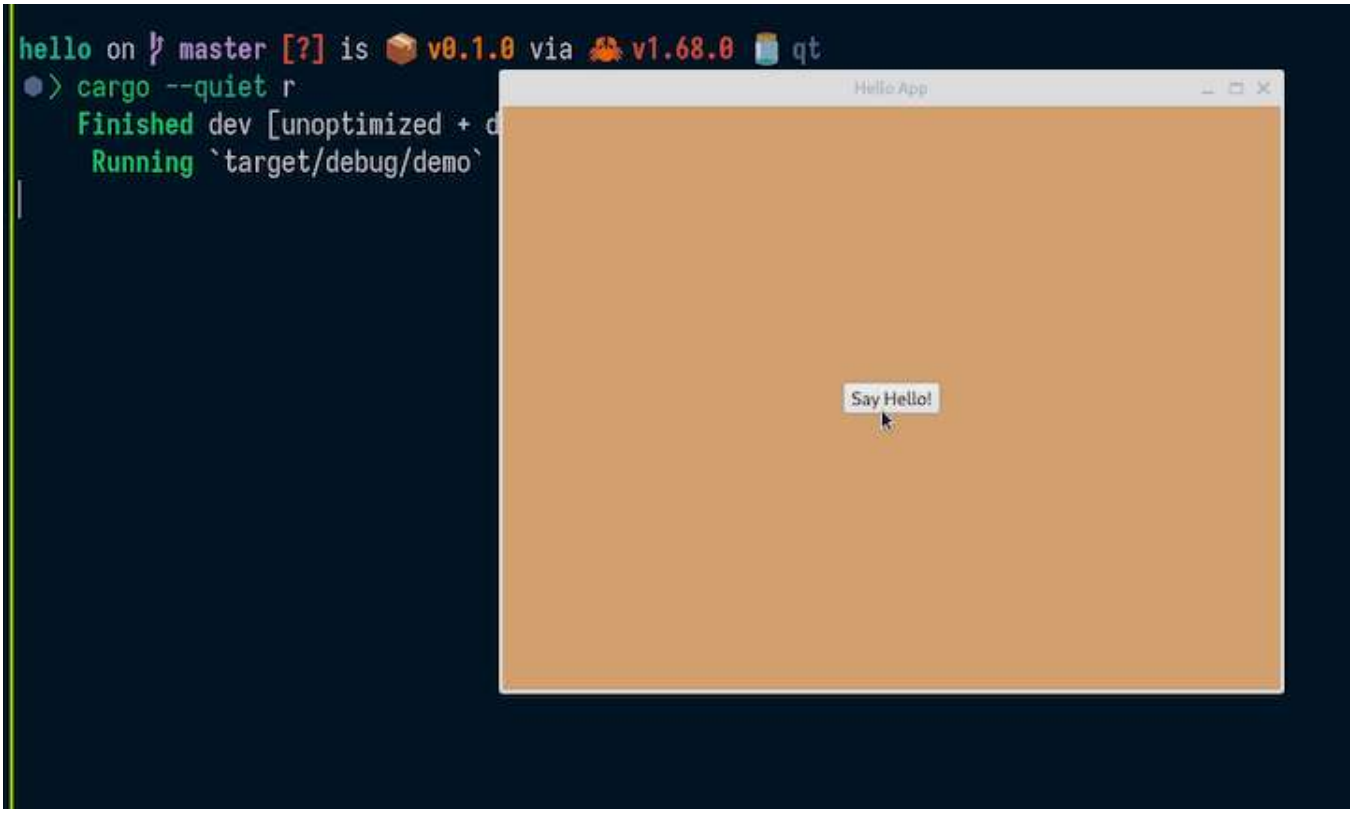
```
⬢ ❯ exa --tree --git-ignore
.
├── qml
│   ├── main.qml
│   └── qml.qrc
├── src
│   ├── cxxqt_object.rs
│   └── main.rs
├── build.rs
├── Cargo.lock
└── Cargo.toml
```

Now, let's use `cargo check` to make sure we have a correct code.

```
# `cargo c` is an alias to `cargo check`
❯ cargo c
  Finished dev [unoptimized + debuginfo] target(s) in 0.04s
```

F

```
⬢ ❯ cargo --quiet r
   Compiling demo v0.1.0
    Finished dev [unoptimized + debuginfo] target(s) in 0.49s
     Running `target/debug/demo`
Hello world!
```



Using the Just task runner (Optional)

A task runner can save you time by automating repetitive development tasks. I recommend using Just. To install Just, use the following command:

```
cargo install just
```

Just is similar to Make, but without Make's complexity and idiosyncrasies.

Simply add the `justfile` file into your root directory and put your repetitive tasks th

```
#!/usr/bin/env -S just --justfile

alias d := dev
alias r := run
alias f := fmt
alias l := lint
alias t := test

# List available commands.
_default:
    just --list --unsorted

# Develop the app.
dev:
    cargo watch -x 'clippy --locked --all-targets --all-features'

# Develop the app.
run:
    touch qml/qml.qrc && cargo run

# Format the codebase.
fmt:
    cargo fmt --all

# Check if the codebase is properly formatted.
fmt-check:
    cargo fmt --all -- --check

# Lint the codebase.
lint:
    cargo clippy --locked --all-targets --all-features

# Test the codebase.
```

```
comply: fmt lint test

# Check if the repository complies with the rules and is ready to be
check: fmt-check lint test
```

For more convenience, you can use `j` as an alias for `just` in your shell:

```
alias j='just'
```

Now you can use `j r` for `cargo run` and so on.

Adding encryption

Let's further improve the application by modifying the text input. We'll add some simple encryption using Caesar Cipher encoding.

First, let's rename the application to `"caesar"`:

```
# Cargo.toml

[package]
-name = "demo"
+name = "caesar"
```

Then, we'll add the `encrypt` function:

```
// cxxqt_object.rs

#[cxx_qt::bridge]
mod my_object {
```

```
    }

    #[cxx_qt::qobject(qml_uri = "caesar", qml_version = "1.0")]
    pub struct Rot {
        #[qproperty]
        plain: QString,
        #[qproperty]
        secret: QString,
    }

    impl Default for Rot {
        fn default() -> Self {
            Self {
                plain: QString::from(""),
                secret: QString::from(""),
            }
        }
    }

    impl qobject::Rot {
        #[qinvokable]
        pub fn encrypt(&self, plain: &QString) -> QString {
            let result = format!("{plain} is a secret");
            QString::from(&result)
        }
    }
}
```

Next, we need to import the `Qstring`, because our `encrypt` function accepts `Qstring` as an argument and also has `Qstring` as a return type. The `struct` now contains two fields: `plain` and `secret`. The `encrypt` function is a regular Rust function, but it accepts and returns a type from CXX-Qt: `Qstring`.

Ir

```
// main.qml

import QtQuick.Controls 2.12
import QtQuick.Window 2.12

// This must match the qml_uri and qml_version
// specified with the #[cxx_qt::qobject] macro in Rust.
import caesar 1.0

Window {
    title: qsTr("Caesar")
    visible: true
    height: 480
    width: 640
    color: "#e4af79"

    Rot {
        id: rot
        plain: ""
        secret: ""
    }

    Column {
        anchors.horizontalCenter: parent.horizontalCenter
        anchors.verticalCenter: parent.verticalCenter
        /* space between widget */
        spacing: 10

        Label {
            text: "Keep your secret safe 🔒"
            font.bold: true
        }
```

```
            background: Rectangle {
                implicitWidth: 400
                implicitHeight: 50
                radius: 3
                color:  "#e2e8f0"
                border.color:  "#21be2b"
            }
        }

        Button {
            text: "Encrypt!"
            onClicked: rot.secret = rot.encrypt(rot.plain)
        }

        Label {
            text: rot.secret
        }
    }
}
```

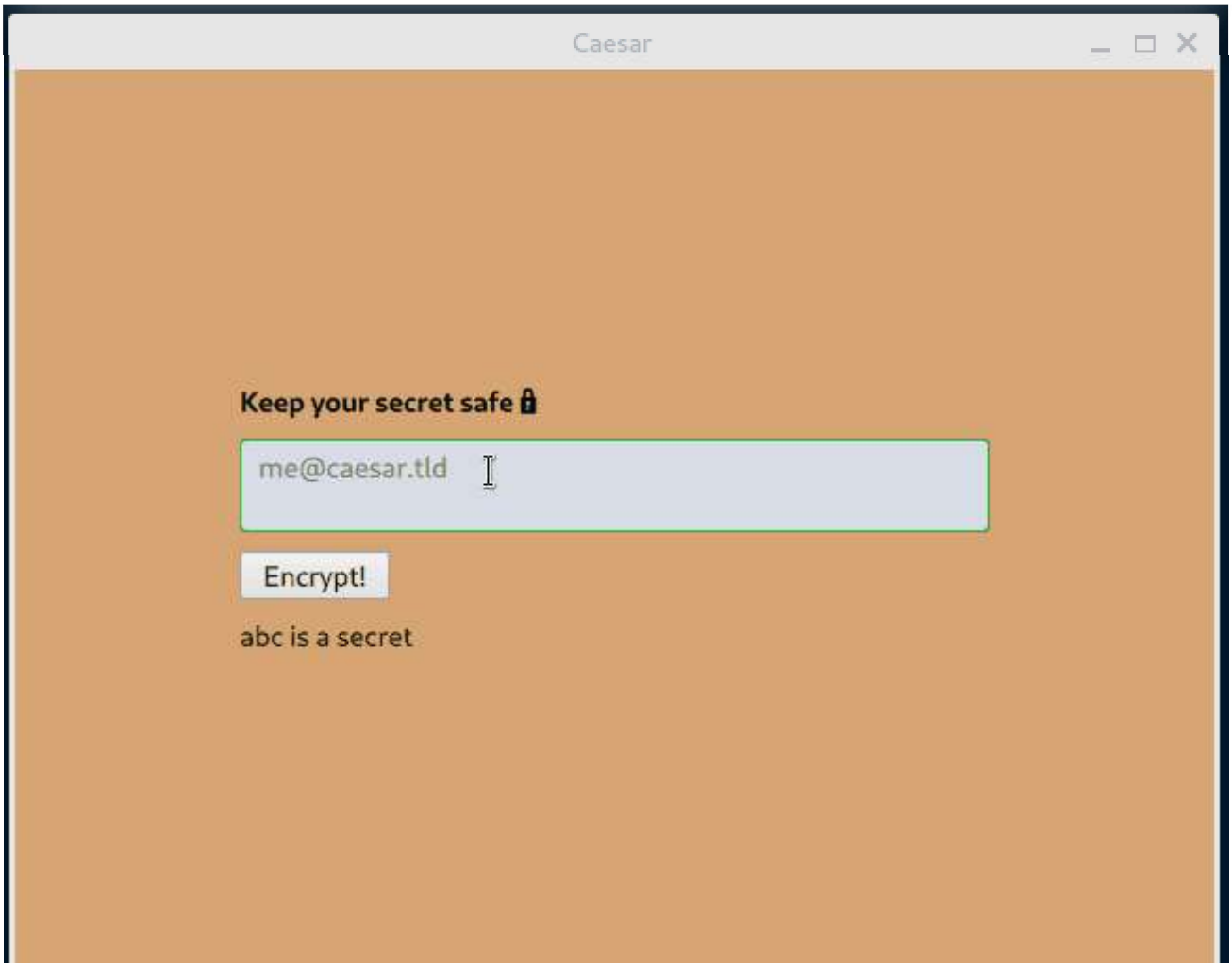
In the `TextArea` we use the `onTextChanged` signal to set the `plain` field of the `Rot struct` to be assigned the value of the `TextArea` input whenever anything changes.

Then, we use the `onClicked` signal in the `Button` component to assign the return value of the `encrypt` function to the `secret` field. Also, we pass the value of `plain` to the `encrypt` function.

Finally, we display the value of the `secret` field in the `Label` component.

Now, let's run the application:

```
⬢ ❯ j r
touch qml/qml.qrc && cargo run
   Compiling caesar v0.1.0
    Finished dev [unoptimized + debuginfo] target(s) in 8.12s
     Running `target/debug/caesar`
```



Encrypting with nrot

One approach for encrypting the user input (secret message) text is to use the nrot third-party library.

First, let's add the dependency:

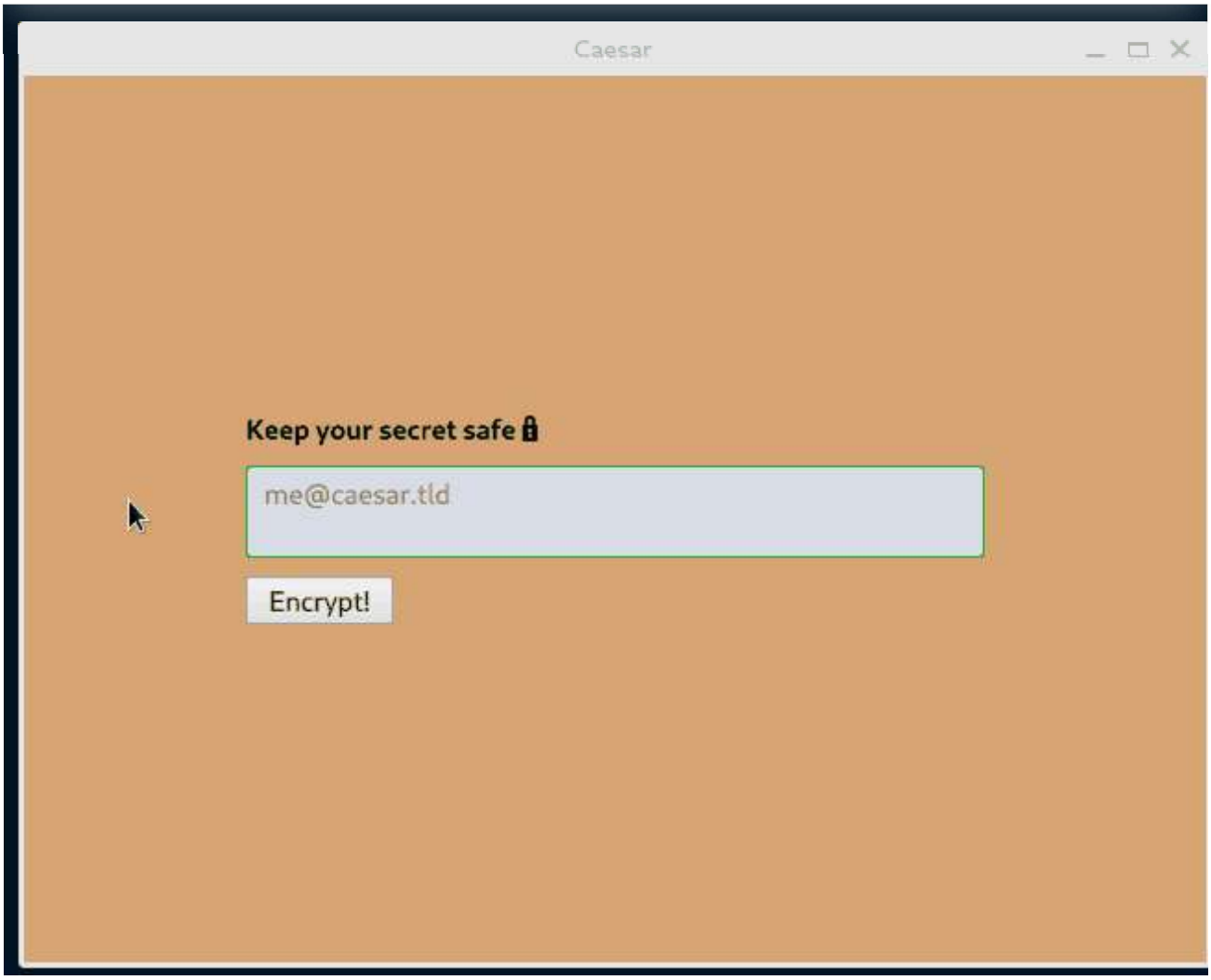
```
# Cargo.toml

+[dependencies]
+nrot = "2.0.0"

+# UI
cxx = "1.0.83"
```

T

```
 mod my_object {

+    use nrot::{rot, rot_letter, Mode};
+
     unsafe extern "C++" {
```

Next, let's use nrot to improve our app's `encrypt` function:

```
        pub fn encrypt(&self, plain: &QString) -> QString {
            let rotation = 13; // common ROT rotation
            let plain = plain.to_string();
            let plain = plain.as_bytes();

            let bytes_result = rot(Mode::Encrypt, plain, rotation);
            let mut secret = format!("{}", String::from_utf8_lossy(&b

            if plain.len() == 1 {
                let byte_result = rot_letter(Mode::Encrypt, plain[0],
                secret = format!("{}", String::from_utf8_lossy(&[byte
            };

            QString::from(&secret)
        }
```



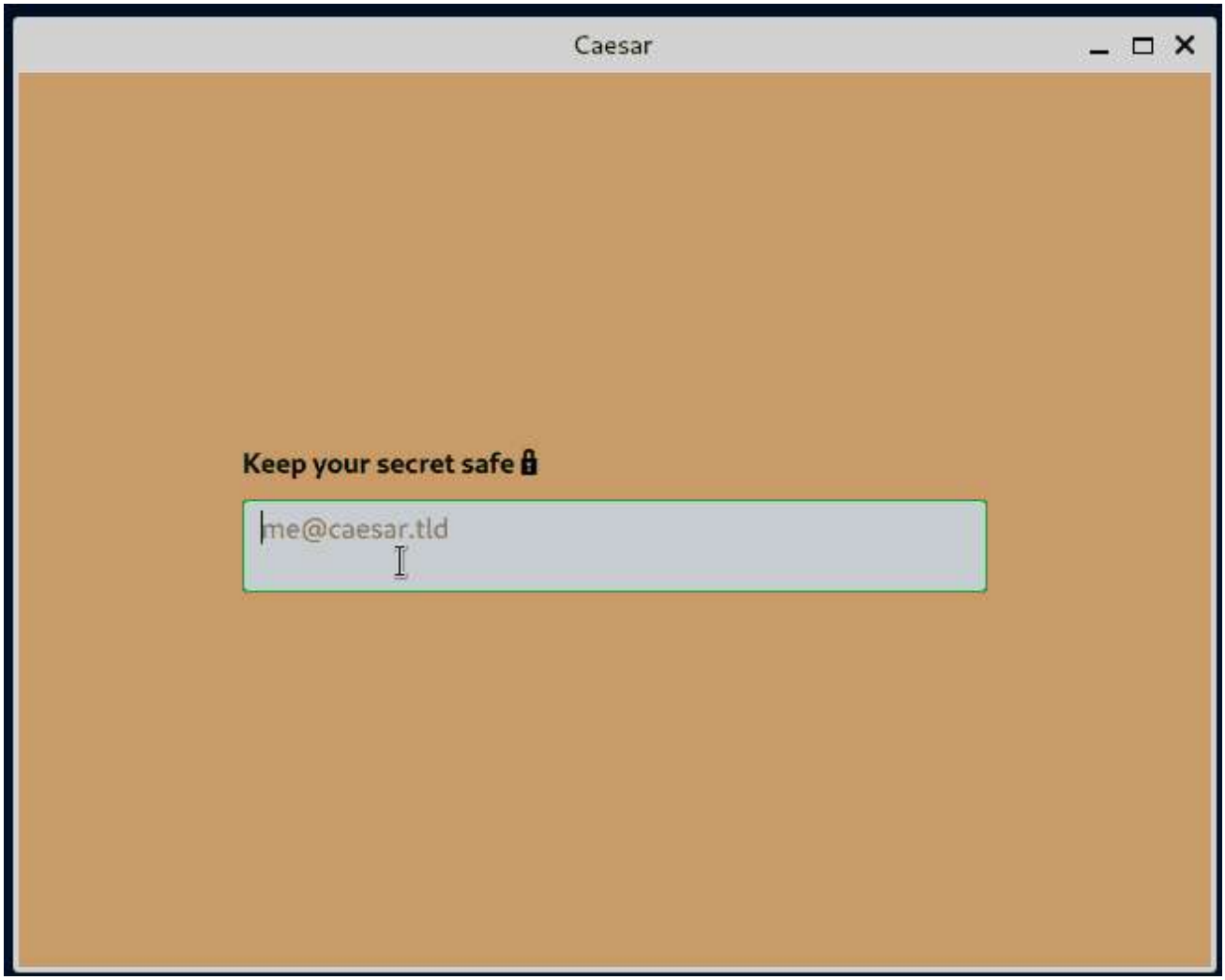
Using on-the-fly encryption

Another approach for encrypting the user input (secret message) text is to use on-the-fly encryption. This will enable us to eliminate the old-fashioned button and actually encrypt the secret message as the user is typing.

First, let's remove the `Button` component and set the value of the `secret` field to update every time the user makes a change to the input field:

```
modified   qml/main.qml
             placeholderText: qsTr("me@caesar.tld")
             text: rot.plain
-            onTextChanged: rot.plain = text
```

```
-            text: "Encrypt!"
-            onClicked: rot.secret = rot.encrypt(rot.plain)
-        }
-
```



Adding decryption

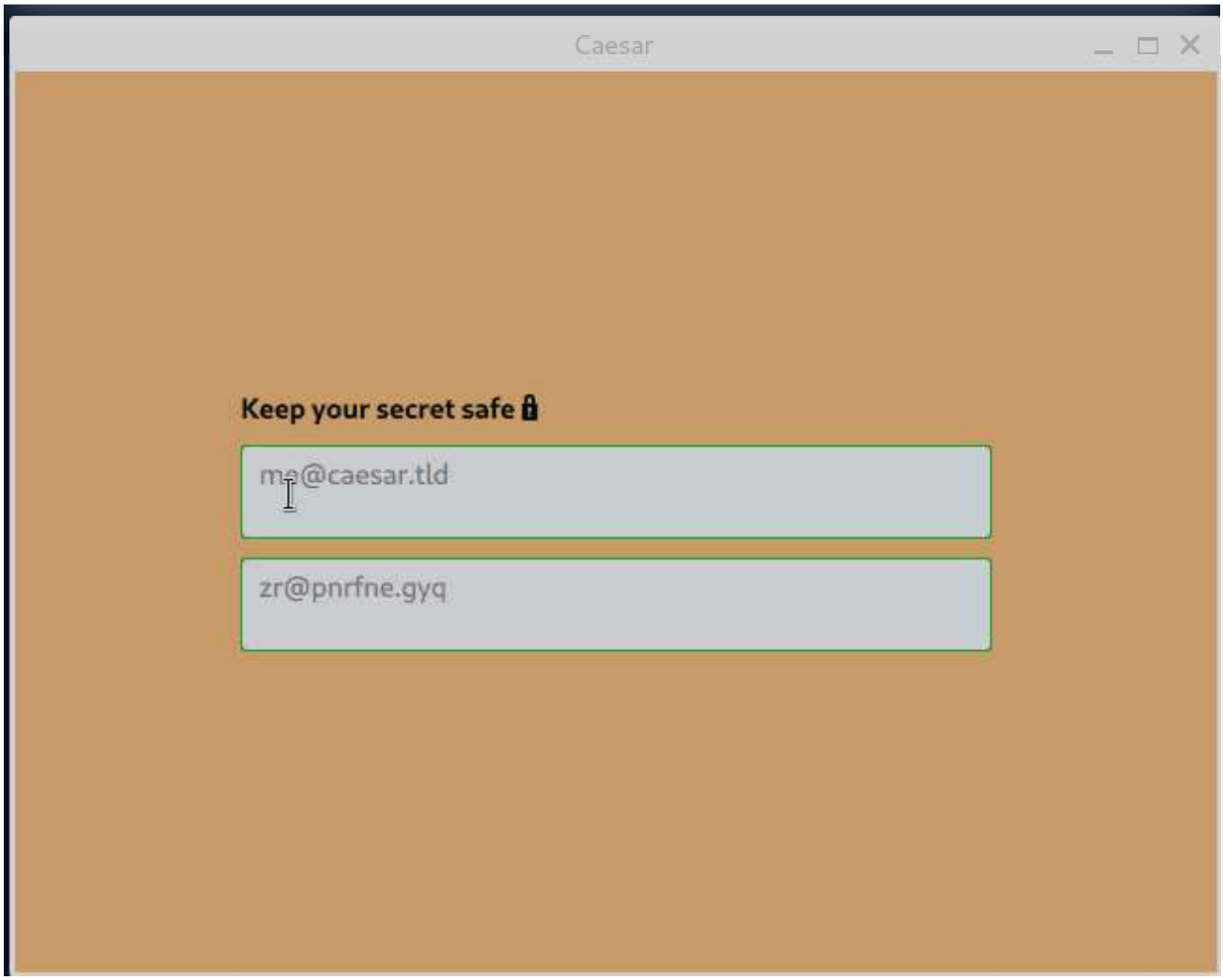
Now, let's add the decryption functionality. We'll use the following `decrypt` function:

```
    #[qinvokable]
    pub fn decrypt(&self, secret: &QString) -> QString {
        let rotation = 13; // common ROT rotation
```

```
        let bytes_result = rot(Mode::Decrypt, secret, rotation);
        let mut plain = format!("{}", String::from_utf8_lossy(&by

        if secret.len() == 1 {
            let byte_result = rot_letter(Mode::Decrypt, secret[0]
            plain = format!("{}", String::from_utf8_lossy(&[byte_
        };

        QString::from(&plain)
    }
```

Then, we'll add the second `TextArea` component for interchangeable input:

```
    TextArea {
        placeholderText: qsTr("zr@pnrfne.gyq")
        text: rot.secret
        onTextChanged: rot.plain = rot.decrypt(text)

        background: Rectangle {
            implicitWidth: 400
            implicitHeight: 50
            radius: 3
            color: "#e2e8f0"
            border.color: "#21be2b"
        }
    }
}
```



Using a custom component to avoid duplication

We have two very similar `TextArea` text fields. To avoid duplication, let's create a custom component.

First, create an `InputArea.qml` file in the `qml` directory with the content of the previous `TextArea` component:

```
// InputArea.qml

import QtQuick 2.12
```

```
    background: Rectangle {
        implicitWidth: 400
        implicitHeight: 50
        radius: 3
        color:  "#e2e8f0"
        border.color: "#21be2b"
    }
}
```

Include the file in the Qt resource:

```
<RCC version="1.0">
    <qresource prefix="/">
        <file>main.qml</file>
+        <file>InputArea.qml</file>
    </qresource>
</RCC>
```

Next, modify the `main.qml` file to use `InputArea`:

```
// main.qml

        InputArea {
            placeholderText: qsTr("me@caesar.tld")
            text: rot.plain
            onTextChanged: rot.secret = rot.encrypt(text)
        }

        InputArea {
            placeholderText: qsTr("zr@pnrfne.gyq")
            text: rot.secret
            onTextChanged: rot.plain = rot.decrypt(text)
        }
```

As a final step, to ensure that we have the correct code in each commit, let's create a GitHub CI workflow:

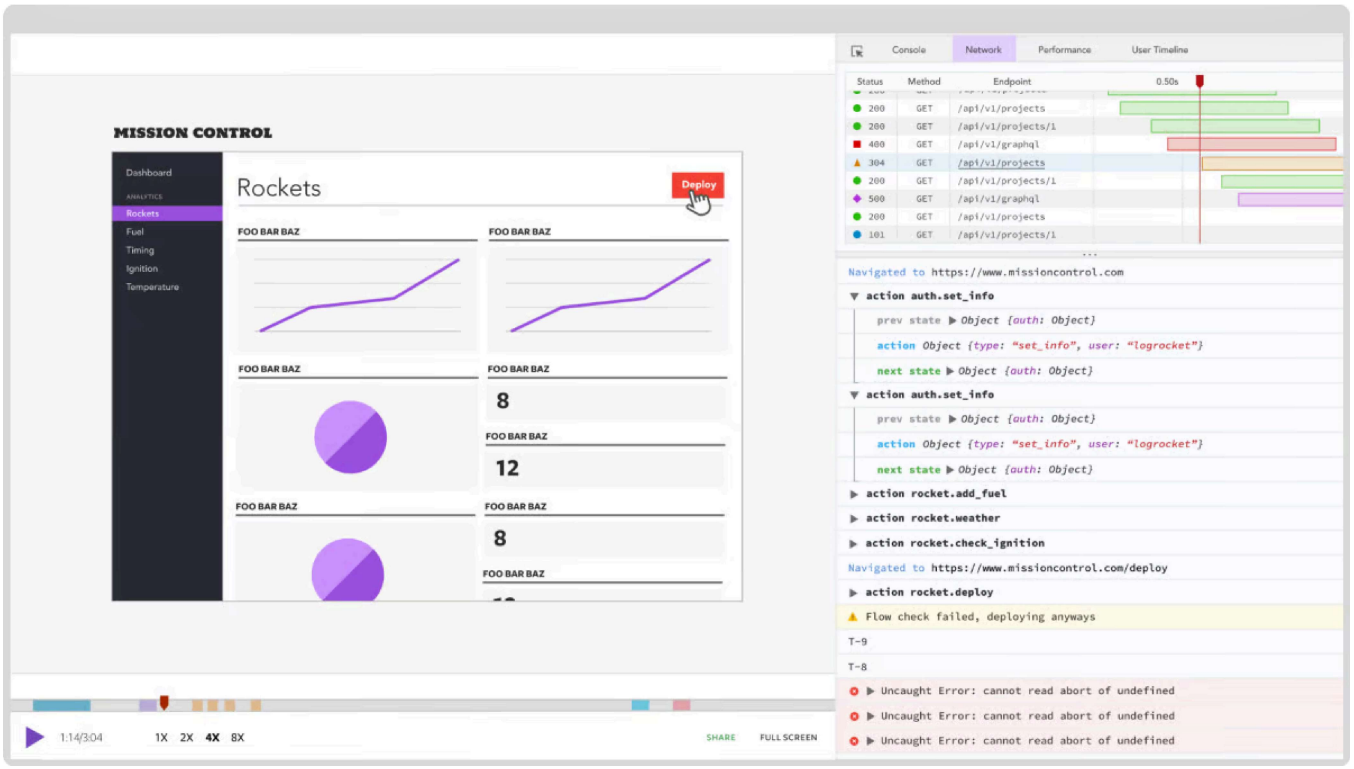
```
name: Caesar

jobs:
  code_quality:
    name: Code quality
    runs-on: ubuntu-22.04

  build:
    name: Build for GNU/Linux
    runs-on: ubuntu-22.04
    strategy:
      fail-fast: false

    steps:
      - name: Checkout source code
        uses: actions/checkout@v3

      - name: Install Rust toolchain
        uses: dtolnay/rust-toolchain@stable
        with:
          target: x86_64-unknown-linux-gnu

      - name: Install Qt
        if: matrix.os == 'ubuntu-22.04'
        run: |
          sudo apt-get update
          sudo apt-get install -y --no-install-recommends --allow-una
            qt6-base-dev
            qt6-declarative-dev
```

Conclusion

In this article, we demonstrated how to build a desktop Qt application using Rust and QML. Along the way, we discussed QObject, Qt signals, and CXX-Qt attribute macros.

If you'd like, you can further improve the demo app by adding more rotation features. Currently, the rotation value is hardcoded to 13. Also, you can play with the current QML user interface to make it fancier. The code for this article's demo application is available on GitHub.

I hope you enjoyed this article. If you have questions, feel free to leave a comment.



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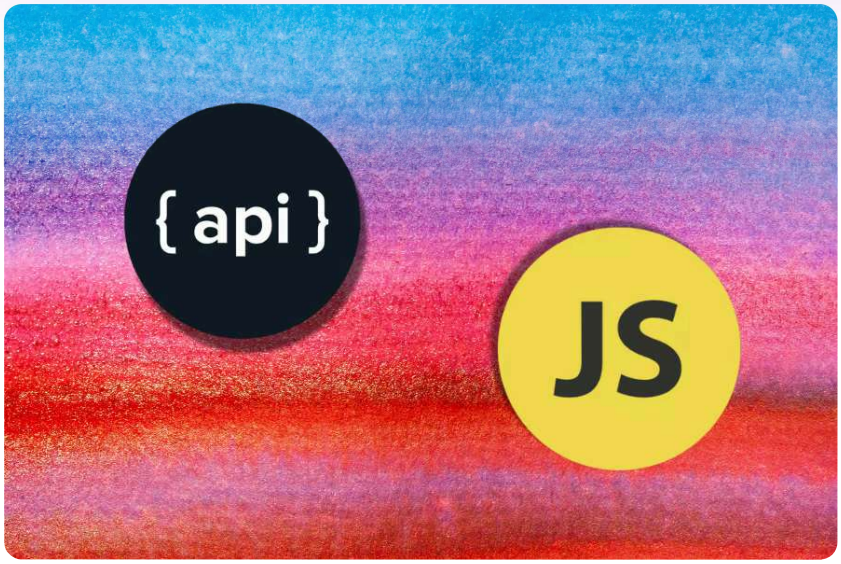

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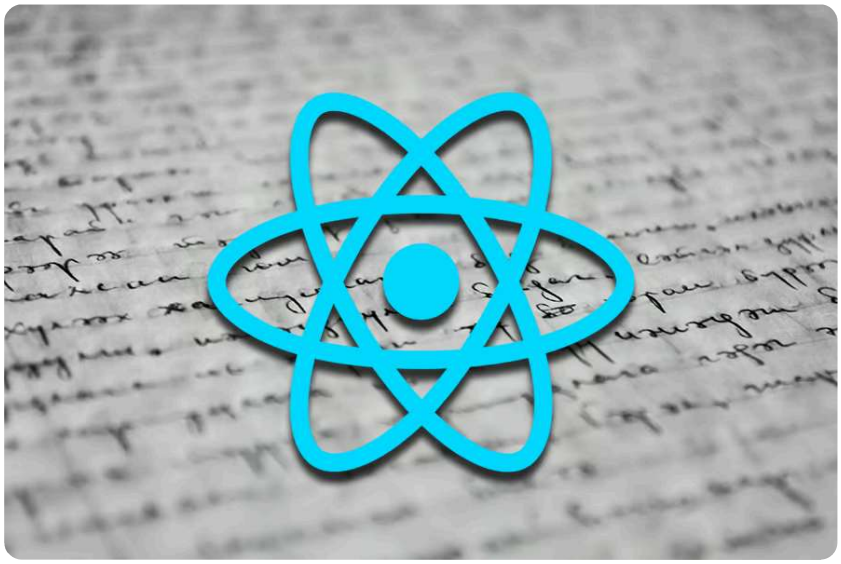

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David says: Reply

April 4, 2024 at 4:42 pm

Hello Azzam,

I tried to start desktop development with RUST an QT on a Windows machine with your tutorial. But I ran into errors.

Let me describe what I've done:

I st
d:\
rus

I installed Qt via the online installer (qt-unified-windows-x64-4.7.0-online.exe), after logging in I chose “Qt 6.7 for Desktop Development”, this installed mingw1120_64, QtCreator and QtDesignStudio on my machine. I set my %path% variable to the mingw-directory “C:\Qt\6.7.0\mingw_64\bin\”, because the qmake.exe is located there:
d:\Eigene Dateien\Rust\Projects\demo>qmake –version
QMake version 3.1
Using Qt version 6.7.0 in C:/Qt/6.7.0/mingw_64/lib
Then I created the source files as you described. I had only one uncertainty with the line
+ mod cxxqt_object;
which was inserted in a second step into main.rs. And what did you mean with “don’t forget to tell Rust if you have a new file”, since I understand the way RUST compiles the files, that it collects the needed files itself according to the directory tree inside the project directory?
I double-checked the final structure and in the project directory I executed
cargo c
and after a lot more than 0.04s I got following error:
Compiling cxx-qt-lib v0.5.3
The following warnings were emitted during compilation:
Warning: ToolExecError: Command “C:\\Program Files (x86)\\Microsoft Visual Studio\\2019\\Enterprise\\VC\\Tools\\MSVC\\14.29.30133\\bin\\HostX64\\x64\\cl.exe” “-nologo” “-MD” “-Z7” “-Brepro” “-I” “d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\include” “-I” “d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\crate” “-I” “C:/Qt/6.7.0/mingw_64/include/QtCore” “-I” “C:/Qt/6.7.0/mingw_64/include/QtGui” “-I” “C:/Qt/6.7.0/mingw_64/include/QtQml” “-I” “C:/Qt/6.7.0/mingw_64/include” “-I” “d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out” “-W4” “/std:c++17” “/Zc:__cplusplus” “/permissive-” “-DCXX_QT_GUI_FEATURE” “-DCXX_QT_QML_FEATURE” “-Fod:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\f9481b67b697aae2-qdatetime.rs.o” “-c” “d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\sources\\cxx-qt-lib\\src\\core\\qdatetime.rs.cc” with arg cc
err

Caused by:
process didn't exit successfully: `d:\Eigene Dateien\Rust\Projects\demo\target\debug\build\cxx-qt-lib-c6d0f297f1efb517\build-script-build` (exit code: 1)
— stderr
CXX include path:
d:\Eigene Dateien\Rust\Projects\demo\target\debug\build\cxx-qt-lib-965a6defbba745e3\out\cxxbridge\include
d:\Eigene Dateien\Rust\Projects\demo\target\debug\build\cxx-qt-lib-965a6defbba745e3\out\cxxbridge\crate
C:/Qt/6.7.0/mingw_64/include/QtCore
C:/Qt/6.7.0/mingw_64/include/QtGui
C:/Qt/6.7.0/mingw_64/include/QtQml
C:/Qt/6.7.0/mingw_64/include
error occurred: Command "C:\\Program Files (x86)\\Microsoft Visual Studio\\2019\\Enterprise\\VC\\Tools\\MSVC\\14.29.30133\\bin\\HostX64\\x64\\cl.exe" "-nologo" "-MD" "-Z7" "-Brepro" "-I" "d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\include" "-I" "d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\crate" "-I" "C:/Qt/6.7.0/mingw_64/include/QtCore" "-I" "C:/Qt/6.7.0/mingw_64/include/QtGui" "-I" "C:/Qt/6.7.0/mingw_64/include/QtQml" "-I" "C:/Qt/6.7.0/mingw_64/include" "-I" "d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out" "-W4" "/std:c++17" "/Zc:__cplusplus" "/permissive-" "-DCXX_QT_GUI_FEATURE" "-DCXX_QT_QML_FEATURE" "-Fod:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\f9481b67b697aae2-qdatetime.rs.o" "-c" "d:\\Eigene Dateien\\Rust\\Projects\\demo\\target\\debug\\build\\cxx-qt-lib-965a6defbba745e3\\out\\cxxbridge\\sources\\cxx-qt-lib\\src\\core\\qdatetime.rs.cc" with args "cl.exe" did not execute successfully (status code exit code: 2).
This shows, there is a problem with the execution of cl.exe of my VS2019- installation. It happens, when the build process is at step 53/57
W
Be
Da

Leave a Reply